#1 Since this algorithm leverages the inductive nature of graphs, we start with the base case: graphs with less than 5 vertices are not planar, for sure.
#2 Check if the number of edges and vertices violates Euler's constraints for planar graphs: if so, this can't be planar.
#3 Finally, if this graph is (isomorphic to) either K_5 or $K_{3,3}$, then for sure it can't be planar.
#4 Cycle through the graph's vertices.
#5 Create a sub-graph by removing v and all the edges adjacent to v.
#6 If the sub-graph so created is not planar, neither is `graph`.
#7 Cycle through all edges in `graph`.
#8 This time create a sub-graph by just removing current edge.
#9 If the resulting sub-graph isn't planar, neither is `graph`.
#10 If we made it all the way to this line, then `graph` is planar.

When it comes to decomposing `G`, therefore, we need to consider two sets of subgraphs:

1. **Induces subgraphs**: all the graphs that could be obtained by individually removing each vertex of `G` (induction rule 1), and all the edges in turn touching the vertex removed;
2. **Spanning subgraphs**: all the graphs that could be obtained by individually removing each edge of `G` (induction rule 2).